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Studierichting: Informatica
Oefeningenreeks 2E nr. 26.5

$$1 + \left(\log_{x-1} x\right)^{-1} = \left(\log_{2x+3} x\right)^{-1} + \log_x x \cdot (\log_2 x)$$

$$\Leftrightarrow 1 + \frac{1}{\log_{x-1} x} = \frac{1}{\log_{2x+3} x} + \frac{\log_{2x+3} 2}{\log_{2x+3} x}$$

$$\Leftrightarrow 1 + \frac{1}{\log_{x-1} x} = \frac{1 + \log_{2x+3} 2}{\log_{2x+3} x}$$

$$\Leftrightarrow 1 + \frac{\log_{2x+3}(x-1)}{\log_{2x+3} x} = \frac{1 + \log_{2x+3} 2}{\log_{2x+3} x}$$

$$\Leftrightarrow 1 = \frac{1 - \log_{2x+3}(x-1) + \log_{2x+3} 2}{\log_{2x+3} x}$$

$$\Leftrightarrow 1 = \frac{\log_{2x+3}(2x+3) - \log_{2x+3}(x-1) + \log_{2x+3} 2}{\log_{2x+3} x}$$

$$\Leftrightarrow 1 = \frac{\log_{4x+6}(2x+3) - \log_{2x+3}(x-1)}{\log_{2x+3} x}$$

$$\Leftrightarrow \log_{2x+3} x = \log_{2x+3}\left(\frac{4x+6}{x-1}\right)$$

$$\Leftrightarrow x = (2x+3)^{\log_{2x+3}\left(\frac{4x+6}{x-1}\right)}$$

$$\Leftrightarrow x = \frac{4x+6}{x-1}$$

$$\Leftrightarrow x(x-1) = 4x+6 \Leftrightarrow x^2 - 5x - 6 = 0$$

$$D = 25 - 4 \cdot 1 \cdot (-6) = 49$$

$$x_1 = \frac{5+7}{2} = 6$$

$x_2 = \dfrac{5-7}{2} = -1$ Kan niet want logaritme van negatief getal kan niet

References